Thakur College of Engineering & Technology

TCET'S NATIONAL LEVEL HACKATHON

HACKANOVA 5.0

Break The Loop, Create the future

Agentic AI

PPT Template

tcet.hackanova@gmail.com

Team Member Details

Team Name	THE STUBBORNS			
Team Members >>	1 (Leader)	2	3	4
Name	Farheen Patel	Daanial Baig	Shaikh Sadaf Noor	Aayush Patil
Institute Name	SIES GST	SIES GST	SIES GST	SIES GST
Email	farheenparvej22@gmail.com	daanialmirza@gmail.com	sadafnoor355@gmail.com	aayushpatil418@gmail.com

HACKANOVA 5.0

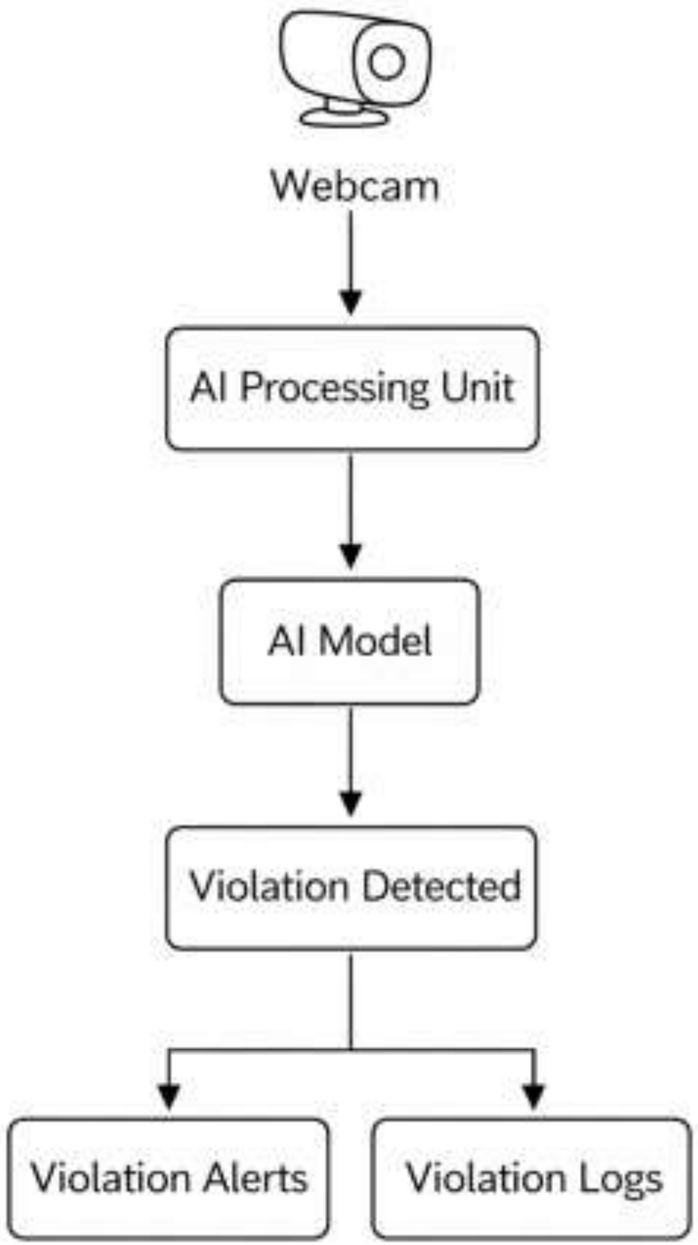
Problem Statement

- Industrial accidents remain a major challenge in factories.
- Workers often ignore PPE (helmet, vest, gloves).
- Manual supervision cannot monitor every worker.
- Factories need automated real-time safety monitoring.
- Goal: AI system that detects safety violations instantly.

HACKANOVA 5.0

Working Principle

- Webcam captures real-time factory video.
- Laptop processes video using AI detection model.
- System identifies missing PPE and unsafe zones.
- Violation triggers buzzer alert and dashboard notification.
- Safety data stored for analytics.



HACKANOVA 5.0

Required Components

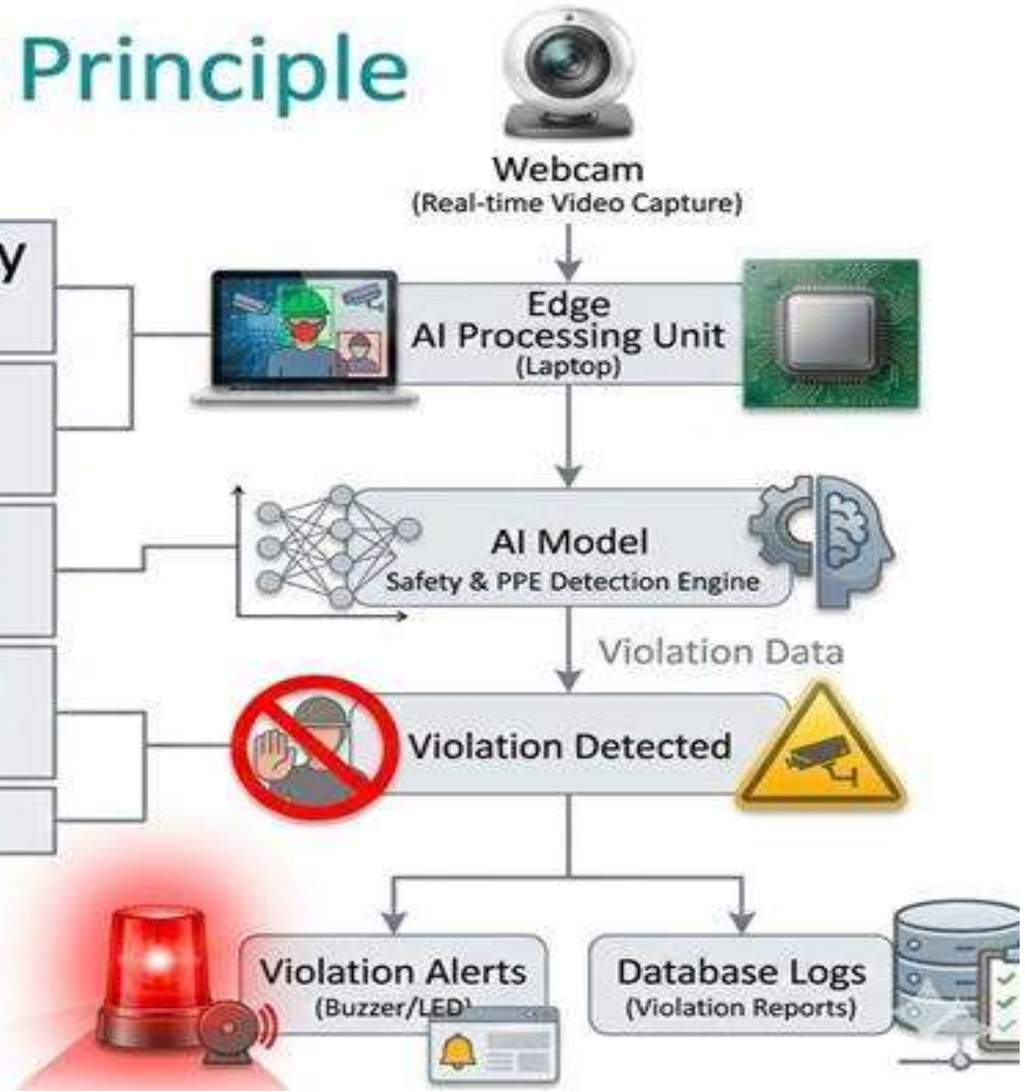
- 1080p HD Webcam
- Laptop (Edge AI Processing Unit)
- Buzzer Alarm
- LED Warning Indicator
- Python, OpenCV, YOLOv8
- Streamlit / Flask Dashboard

HACKANOVA 5.0

Flowchart/Block Diagram/ Workflow Diagram

Working Principle

- Webcam captures real-time factory video.
- Laptop processes video using AI detection model.
- System identifies missing PPE and unsafe zones.
- Violation triggers buzzer alert and dashboard notification.
- Safety data stored for analytics.



HACKANOVA 5.0

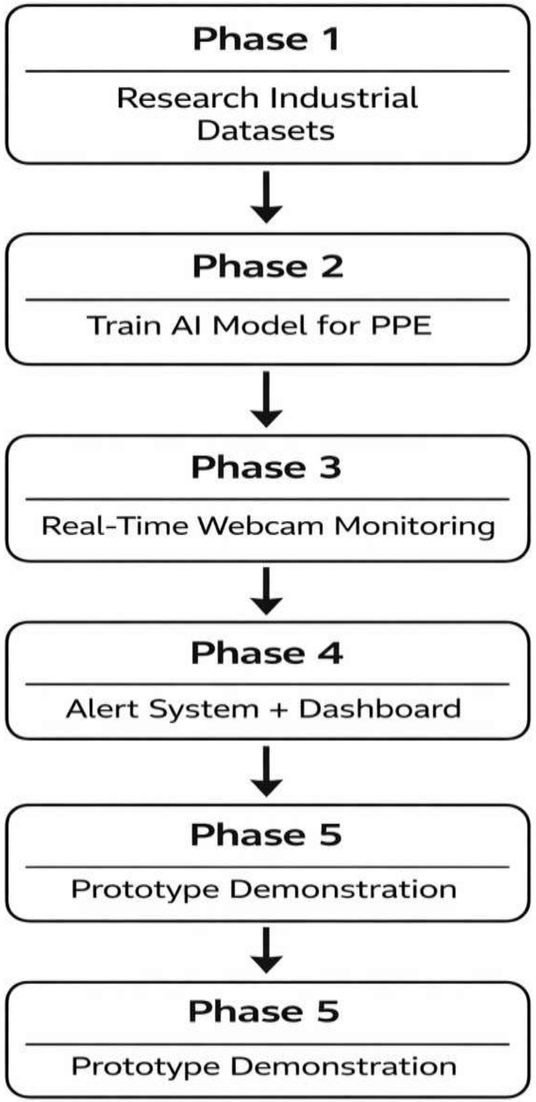
Expected Outcomes

- Real-time worker safety monitoring
- Instant detection of safety violations
- Reduced workplace accidents
- Improved industrial safety compliance
- Scalable Industry 4.0 safety solution

HACKANOVA 5.0

Execution Plan

- **Phase 1:** Research industrial safety datasets
- **Phase 2:** Train AI model for PPE detection
- **Phase 3:** Implement real-time webcam monitoring
- **Phase 4:** Integrate alert system and dashboard
- **Phase 5:** Demonstrate prototype



HACKANOVA 5.0

Potential Problems and Limitations

Challenges	Solutions / Future Improvements
Lighting variations affecting detection accuracy	✓ Train model with diverse lighting datasets and apply image preprocessing techniques
Workers blocking camera view (**occlusion**)	✓ Use multiple camera angles or improved tracking algorithms
Limited processing power on edge devices (laptop)	✓ Optimize AI model and use lightweight models like YOLOV8-nano
Real-time detection latency in busy environments	✓ Use GPU acceleration or dedicated edge AI devices
System scalability for large factories	✓ Integrate multi-camera monitoring system
Limited environmental sensing	✓ Combine with IoT sensors (gas, temperature, motion)

HACKANOVA 5.0